

PROJECT DOCUMENTATION

Posture Correction Alert System Using Arduino and Flex Sensor

Group Number: 30

Group Members:

- | | |
|--------------------------------|---------|
| • PARKER MCKEOWN EWURABA ESEBA | 7281823 |
| • KWAME OWUSU NKETIA | 7281223 |
| • YAW OSEI-SEMPREMOO | 7280423 |
| • TENNYSON JOOJO KAKRA | 7283523 |
| • OSMAN ZAINAB TWENEBOAH | 7280523 |
| • GEORGE SARFO TERRY | 7283623 |
| • PEPRAH PAPA KWASI TAKYI | 7281923 |
| • OPPONG MARVIN | 7279923 |
| • KELVIN SENA WEDE | 7284323 |
-

1. Introduction

Maintaining proper posture is essential for preventing long-term health issues such as back pain, spinal misalignment, and fatigue. However, many individuals develop poor posture habits due to prolonged sitting and lack of awareness.

This project presents a **Posture Correction System** that monitors the user's posture in real time and provides immediate feedback when poor posture is detected. The system uses a flex sensor to detect bending and an Arduino microcontroller to process the data and trigger alerts.

2. Objectives

The main objectives of this project are:

- To design a system that detects poor posture using a flex sensor
 - To provide real-time feedback using a buzzer and LED
 - To implement sensor calibration for accurate measurements
 - To develop a low-cost and simple posture monitoring solution
-

3. Components Used

Component	Description
Arduino Uno	Microcontroller used to process input and control outputs
Flex Sensor	Detects bending by changing resistance
Resistors	Used for voltage divider and current limiting
LED	Provides visual alert
Buzzer	Provides sound alert
Breadboard	Used for circuit assembly
Jumper Wires	Used for connections

4. System Overview

The system works by detecting the bending of a flex sensor attached to the user's back or posture-support structure.

Working Principle

- The flex sensor changes resistance when bent
 - This resistance change is converted into voltage using a voltage divider
 - The Arduino reads this voltage as an analog signal
 - The signal is converted into an angle value
 - If the angle exceeds a preset threshold, alerts are triggered
-

5. Circuit Description

- The flex sensor is connected in a **voltage divider configuration** with a resistor
 - The midpoint of the divider is connected to **Analog Pin A0**
 - The LED and buzzer are connected to **digital output pins**
 - A common ground is shared across all components
 - The Arduino supplies **5V power** to the circuit
-

6. Software Design

The system is programmed using the Arduino IDE.

Key Functions Used

- `analogRead()` → Reads sensor value
 - `map()` → Converts sensor value to angle
 - `digitalWrite()` → Controls LED and buzzer
-

7. Code Explanation

```
int sensorPin = A0;
int ledPin = 4;
int buzzer = 4;

int sensorValue = 0;
int angle = 0;

int minVal = 256; // straight
int maxVal = 59;  // bent

void setup() {
  Serial.begin(9600);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzer, OUTPUT);
}

void loop() {
  sensorValue = analogRead(sensorPin);

  angle = map(sensorValue, minVal, maxVal, 0, 180);

  Serial.print("Sensor: ");
  Serial.print(sensorValue);
  Serial.print("  Angle: ");
  Serial.println(angle);

  if (angle > 20) {
    digitalWrite(ledPin, HIGH);
    digitalWrite(buzzer, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
    digitalWrite(buzzer, LOW);
  }

  delay(200);
}
```

Explanation

- The sensor value is read from analog pin A0
- The value is mapped to an angle range (0–180°)
- If the angle exceeds 20°, poor posture is detected
- The LED and buzzer are activated

8. Calibration

Calibration is necessary because the flex sensor does not directly output angles.

Steps Taken

1. Measured sensor value when straight → **256**
2. Measured sensor value when bent → **59**
3. Used these values in the `map()` function

This ensures accurate posture detection.

9. Testing and Results

Testing Process

- Observed sensor values using Serial Monitor
- Adjusted threshold values
- Verified LED and buzzer activation

Results

- The system successfully detects posture changes
 - Alerts are triggered when bending exceeds threshold
 - System responds in real time
-

10. Challenges Faced

- Incorrect sensor readings initially
- Difficulty matching sensor values to angles
- Pin conflicts in early code versions
- Noise and fluctuations in sensor data

Solutions

- Used calibration to fix value range
- Used Serial Monitor for debugging
- Adjusted threshold values for stability

11. Applications

- Posture monitoring for students
- Office workers sitting for long periods
- Physiotherapy and rehabilitation
- Wearable health devices

12. Advantages

- Low cost
- Simple design
- Real-time feedback
- Easy to implement

13. Limitations

- Accuracy depends on calibration
- Limited to single sensor detection
- Not yet wearable in current form
- No data storage or tracking

14. Future Improvements

- Add Bluetooth/WiFi for remote monitoring
- Use multiple sensors for better accuracy
- Replace buzzer with vibration motor
- Develop wearable version (belt or strap)
- Add mobile app integration

15. Conclusion

This project successfully demonstrates how a flex sensor and Arduino can be used to detect poor posture and provide immediate feedback.

It highlights the practical application of embedded systems in solving real-world health problems. With further improvements, this system can be developed into a fully wearable and smart posture correction device.
